AIMS5701: Fundamentals in AI

Homework 3: Bayes Nets II

Deadline: 23:59, October 19, 2026

Problems Only | Total: 100 marks

Name: _____ **Student ID:** _____

Important Regulations and Policies

Submission Guidelines

- Please submit your declaration and solutions on Blackboard (under the "course contents" tab).
- Your submission must be a **single PDF file**. Submissions in other formats (e.g., image files, Word documents) will not be graded.
- Please write your name and student ID on the first page of your solutions.
- For the late submission policy, please refer to regulations posted on Blackboard.

Collaboration Policy

- You are allowed to discuss the problems with your classmates to better understand the concepts.
- However, you **MUST** write up your own solutions independently.
- You **MUST NOT** copy from your classmates or from any other material. Do not share your final written solutions with others.

Policy on AI Tools (e.g., ChatGPT, Copilot, Gemini)

- Our policy is "Approach 2 – Use only with prior permission." For this assignment, you are permitted to use AI tools to help you understand concepts.
- **Acknowledgement is mandatory.** If you use an AI tool for any part of a problem, you **MUST** attach the full, unedited transcript of your interaction with the tool for that problem. This includes the prompts you used and the original answer generated by the AI. This should be attached at the end of your submission. Failure to do so will be considered a violation of academic integrity.

Instructions

There are **six problems**. Show all sums, products, intermediate factors, and normalization steps where requested. In Problems 1–5, all variables are binary, taking values in $\{0, 1\}$; omitted probabilities are complements. Each problem's evidence applies only to that problem. Keep unrounded intermediate values and round final probabilities to **three decimal places**; round ESS to **two decimal places**.

Problem 1: Variable Elimination with Evidence [20 Marks]

A monitoring system has four variables: high load H, vibration V, a fault flag F, and an alarm A. Its Bayes net has edges

$$H \to V, \qquad V \to F, \qquad H \to A, \qquad V \to A.$$

The conditional probability tables (CPTs) are:

$P(H{=}1) = 0.4,$

H	0	1
$P(V{=}1 \mid H)$	0.2	0.7

V	0	1
$P(F{=}1 \mid V)$	0.1	0.8

(H, V)	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$P(A{=}1 \mid H, V)$	0.05	0.50	0.60	0.90

The evidence is $A{=}1$. The query is $P(F \mid A{=}1)$.

(a) Instantiate the evidence in the relevant factors and list the resulting factor set, including each factor's scope. **[5 Marks]**

(b) Use variable elimination with elimination order H, then V, to obtain an *unnormalized* distribution over F. Show every intermediate join and elimination, including the factor scopes and resulting tables. **[9 Marks]**

(c) Normalize your result and report $P(F{=}1 \mid A{=}1)$ to three decimal places. **[6 Marks]**

Problem 2: Enumeration with a Common Parent [15 Marks]

A product defect D can cause a positive automated test T and a positive visual inspection I. The graph is $D \to T$, $D \to I$, with CPTs:

$P(D{=}1) = 0.25,$

D	0	1
$P(T{=}1 \mid D)$	0.15	0.85

D	0	1
$P(I{=}1 \mid D)$	0.20	0.80

Compute the following, showing sums and products. Report each answer to three decimal places.

(a) $P(D{=}1 \mid T{=}1, I{=}0)$. **[8 Marks]**

(b) $P(T{=}1 \mid I{=}1)$, by marginalizing over D. **[7 Marks]**

Problem 3: Prior and Rejection Sampling [15 Marks]

A data center has four variables: high activity H, a backup job B, a large upload U, and network congestion N. The graph is

$$H \to B, \qquad H \to U, \qquad B \to N, \qquad U \to N.$$

Use the following CPTs for **Problems 3–5**:

$P(H{=}1) = 0.6,$

H	0	1
$P(B{=}1 \mid H)$	0.2	0.6

H	0	1
$P(U{=}1 \mid H)$	0.25	0.75

(B,U)	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
$P(N{=}1 \mid B,U)$	0.00	0.80	0.70	0.95

(a) Compute $P(N{=}1)$ exactly by marginalization. **[6 Marks]**

(b) To estimate $P(B \mid N{=}1)$ by rejection sampling, complete assignments are drawn from the prior and retained only if $N{=}1$. Give the acceptance probability to three decimal places. **[3 Marks]**

(c) Suppose 12 samples have been *accepted*: 5 have $B{=}1$ and 7 have $B{=}0$. Give the estimate $\widehat{P}(B{=}1 \mid N{=}1)$. **[6 Marks]**

Problem 4: Likelihood Weighting and ESS [15 Marks]

Use the network and CPTs from Problem 3. The evidence is $H{=}1$ and $N{=}1$. Likelihood weighting fixes these evidence variables and samples only B and U. Each sample has weight

$$w = P(H{=}1)\, P(N{=}1 \mid B, U).$$

Five weighted samples are provided:

Sample i	1	2	3	4	5
B_i	1	0	1	0	1
U_i	0	1	1	0	1
w_i	0.420	0.480	0.570	0.000	0.570

(a) Compute the likelihood-weighting estimate

$$\widehat{P}(B{=}1 \mid H{=}1, N{=}1) = \frac{\sum_i w_i\, \mathbf{1}[B_i{=}1]}{\sum_i w_i}$$

to three decimal places. **[9 Marks]**

(b) Compute the effective sample size $\mathrm{ESS} = \dfrac{(\sum_i w_i)^2}{\sum_i w_i^2}$ to two decimal places. **[6 Marks]**

Problem 5: Gibbs Sampling via Markov Blankets [20 Marks]

Use the network and CPTs from Problem 3. In this problem, the *only fixed evidence* is $U{=}1$. The Markov blanket of B is $\{H, U, N\}$, and

$$P(B \mid H, U, N) \propto P(B \mid H)\, P(N \mid B, U).$$

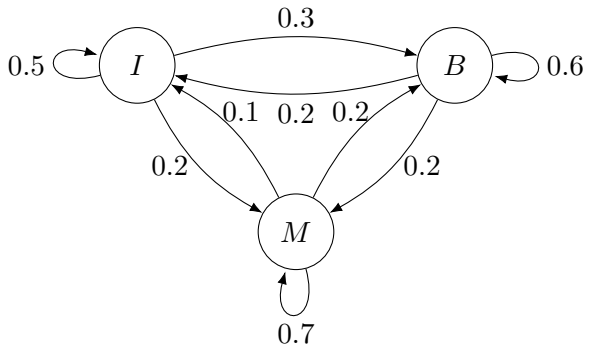
(a) Derive a normalized expression for $P(B{=}1 \mid H{=}0, U{=}1, N{=}1)$ from the CPTs and evaluate it to three decimal places. **[8 Marks]**

(b) The current Gibbs state is $(H{=}0, B{=}0, U{=}1, N{=}1)$. When B is resampled, what is the probability that it flips to 1? Use your result from (a). **[4 Marks]**

(c) Compute $P(B{=}1 \mid U{=}1)$ and $P(B{=}0 \mid U{=}1)$ exactly by marginalization. Show the sums, report both values to three decimal places, and state which is larger. **[8 Marks]**

Problem 6: State-Transition Probabilities [15 Marks]

A server follows a time-homogeneous Markov chain with states idle I, busy B, and maintenance M. The directed graph gives all transition probabilities.



Assume $X_0 = I$. Report exact values or three-decimal approximations.

(a) Give $P(X_1{=}M \mid X_0{=}I)$. **[4 Marks]**

(b) Compute $P(X_2{=}M \mid X_0{=}I)$, using $\mathbf{P}^2$ or a sum over the intermediate state X_1. **[6 Marks]**

(c) Compute $P(X_3{=}B \mid X_0{=}I)$, using $\mathbf{P}^3$ or stepwise summation. **[5 Marks]**